ecbench statistics v2 (PR #2442): before/after evidence

Date 2026-11-05. Study directory: `research/ecbench_statistics_v2_20261105/`. PDF: `REPORT.pdf`. Scope: what PR #2442 (merge `7edc35a9fb`: linearised t(K-1) ops interval, ns/GAE wall statistic, censored timeouts/exhausted runs, tail flags, A/A bias correction) changed in the numbers `ecbench compare` and `ecbench table` print for sessions that were already committed. #2442 landed without this record; this is it. Nothing here is a new run, a speed result or an IC leaderboard row.

Method (all numbers below are measured from the files named; derived ones say how)

- Old statistic: `ecbench` built from `7edc35a9fb^1` = `b58840ef4f` (pre-merge, schema `ecbench.comparison/v1`) in a scratch worktree (the build needed five `include_str!` data files added to the sparse checkout; nothing else changed). Binary sha256 prefix `d2dc241024a419e7`.
- New statistic: `ecbench` built from this branch's base `a4dbbd282d` (current main, `ecbench.comparison/v2`), prefix `e5273d184198c976`.
- Both run `compare --json` on the same committed `records.jsonl` (no rerun of any session, default seed 20261002, 4000 resamples): `extract.sh` -> `data/raw/{old,new}/*.json`, `data/comparisons.tsv`, `data/tables/{old,new}.*.md`. Re-running `extract.sh` reproduced the raw JSON byte-for-byte.
- Sanity check (verified): for every pair, the old ratio equals the new `verified_only_ratio` (e.g. 1.3509 for sub8-da-sat), so the two binaries agree on the point estimate over verified pairs; every difference below is the interval, the censoring or the flags.
- Sessions: yield sweep koblitz `ECBS1h0d1ce1d04e54` (16 workloads, 7 and 4 timeouts on the two `da-sat` arms); `koblitz-followup-3` `ECBS1hb0cb85a662d8` (16 workloads, 24 of 48 `ic-sub6-mitm` runs exhausted, has an A/A arm); ghs `panel-01` `ECBS1h5a490d269b02` (40 workloads incl. 13 exhausted `rho-frob`); sat_pdp `pilot` `ECBS1h2cae65a6c917` (2 workloads, timeouts, K = 2); yield sweep prime `ECBS1h548d49a71881` (12 pairs, clean). Paths in `data/pairs.txt`.

1. Ops ratio: interval width and censoring ([fig 1])

Ratio = sum S_B / sum S_A in ecbench.gae; intervals 95%. Old = percentile cluster bootstrap over verified pairs; new = linearised t(K-1) with censored pairs included.

session: B / A	old ratio [interval]	new ratio [interval]	new status
yield koblitz: sub8-mitm / rho-strong	45.22 [37.34, 49.17]	45.22 [39.74, 51.44]	complete, K=16
yield koblitz: sub8-da-f4 / rho-strong	1.796 [1.406, 2.625]	1.796 [1.289, 2.502]	complete
followup3: ic-sub8-mitm / rho-strong	76.97 [55.96, 100.0]	76.97 [57.01, 103.9]	complete
ghs: rho-neg / bsgs-neg	4.185 [3.728, 4.647]	4.185 [3.725, 4.702]	complete, K=40
prime: abs16-mitm / rho-neg	9.398 [6.161, 14.15]	9.398 [5.702, 15.49]	K=12
prime: abs64-subtract / rho-neg	174.1 [98.3, 300.5]	174.1 [89.65, 338.0]	K=12
sat pilot: ic-mitm / rho-strong	102.3 [99.07, 105.8]	102.3 [67.33, 155.5]	K=2, df 1, `indicative`
yield koblitz: sub8-da-sat / rho-strong	1.351 [1.202, 1.547] (9 of 16 pairs)	1.429 [1.241, 1.646]	lower bound, 7 censored
yield koblitz: sub6-da-sat / rho-strong	1.892 [1.259, 3.217] (12 pairs)	2.486 [1.442, 4.288]	lower bound, 4 censored
followup3: ic-sub6-mitm / rho-strong	6.981 [6.252, 7.730] (24 of 48)	**648.2 [419.6, 1001]**	lower bound, 24 exhausted pairs censored
ghs: rho-frob / bsgs-neg	3.512 [2.779, 4.319] (27 of 40)	212.0 [123.2, 365.1]	lower bound, 13 exhausted pairs censored
sat pilot: ic-buchberger / rho-strong	no ratio (0 verified pairs)	bound 0 [0, 0], 2 censored	uninformative: no verified B run to convert wall time to cost

Reading (widths hi - lo derived from the table above): on uncensored rows the new interval is 13.1x wider at K = 2 (88.17 against 6.73, the old one being the percentile bootstrap that under-covers there), 1.23x wider at K = 12 (abs16 9.788 vs 7.989; abs64 248.4 vs 202.2), 1.06x at K = 16 for followup3 sub8-mitm (46.89 vs 44.04) and 1.00x for yield sub8-mitm (11.70 vs 11.83) and sub8-da-f4 (1.213 vs 1.219), and 1.06x at K = 40 (ghs rho-neg 0.977 vs 0.919). Censoring is the large effect: dropping exhausted/timed-out runs understated `ic-sub6-mitm/rho-strong` by 648.2/6.981 = 92.9x and `rho-frob/bsgs-neg` by 212.0/3.512 = 60.4x (point estimates; both are lower bounds, so the true ratio is higher still). The "648" figure of the PR stands: it is this row, 648.2 [419.6, 1001], bound `lower`, from the same records. The yield-sweep bound in the PR body was withdrawn and is not used here.

2. Coverage simulation ([fig 2])

The PR's coverage claims are backed by the Rust tests in `src/cryptanalysis/ecbench/stats.rs` (the `coverage.py` they cite is not committed, and Python is not allowed here). Reproduced (`data/stats_tests.log`): all 8 `stats::tests` pass, and the committed test prints linearised coverage 0.9573 / 0.9480 / 0.9470 at K = 2 / 4 / 8 (ratio scale) and 0.9557 / 0.9460 / 0.9500 (log scale), matching README section 8 (0.957, 0.948, 0.947 / 0.956, 0.946, 0.950).

This study adds one ignored test, `coverage_table_for_report` (same cluster model, 3 rounds per workload, 2000 replicates per K, 400 bootstrap resamples; Monte-Carlo s.e. about 0.005), comparing the new interval with the old one at K = 2, 3, 4, 8 (`data/coverage.tsv`):

K	linearised t, log scale: coverage / median width	percentile cluster bootstrap: coverage / median width
2	0.9395 / 16.64	0.8580 / 1.631
3	0.9585 / 2.819	0.9055 / 1.518
4	0.9515 / 1.802	0.9275 / 1.323
8	0.9500 / 0.983	0.9620 / 0.980

Differences from the README text, stated as measured: the bootstrap's K = 3 coverage here is 0.906 (README says 92-93% "at three or four"; K = 4 gives 0.928); and at K = 2 the linearised log-scale coverage is 0.940 in this run against 0.956 in the committed test (different seed stream and 2000 against 3000 replicates; a 0.016 gap is about 3 Monte-Carlo s.e., so K = 2 is slightly below nominal on the log scale in this run). The price of honest coverage at small K is width: 10x the bootstrap at K = 2, the same at K = 8. The wall simulation quoted in README section 8 (5 workloads x 3 rounds, "deep dive") is not committed and is **not reproduced** here; it is not relied on.

3. Wall statistic and the A/A gate ([fig 4])

v1 headline: median per-pair solve-wall ratio, whose bootstrap interval was often a point (e.g. [350714.0, 350714.0] for sub8-da-sat, or the median at an edge: [40.28, 55.03] with median 55.03 for followup3 sub8-mitm). v2 headline: ratio of ns per GAE with a linearised t interval over pairs. These are different estimands (total wall versus cost per operation) and often disagree in direction: yield koblitz sub8-mitm/rho-strong is 20.26 (v1 median) but 0.627 [0.489, 0.804] ns/GAE (v2), because the arm does 45x the operations at lower cost per operation. The `da-sat` arms' ns/GAE (5.1e7 and 8.8e6 ns per GAE) reflect unpriced SAT-solver work, and the arms are flagged as upper bounds.

A/A gate, followup3 (A/A pair `rho-strong`/`rho-strong-aa`, "includes arm A"): bias 1.087 [0.967, 1.223]. For `ic-sub8-mitm` the ns/GAE ratio 1.361 [1.229, 1.506] corrects to 1.251 [1.075, 1.456] and stays outside noise (true; v1 rule also true). For `ic-sub6-mitm` 1.248 [1.145, 1.360] corrects to 1.147 [0.997, 1.321]: `outside_noise` false, where the legacy v1 rule says true. This is the one gating flip in the sample.

4. Flags ([fig 3])

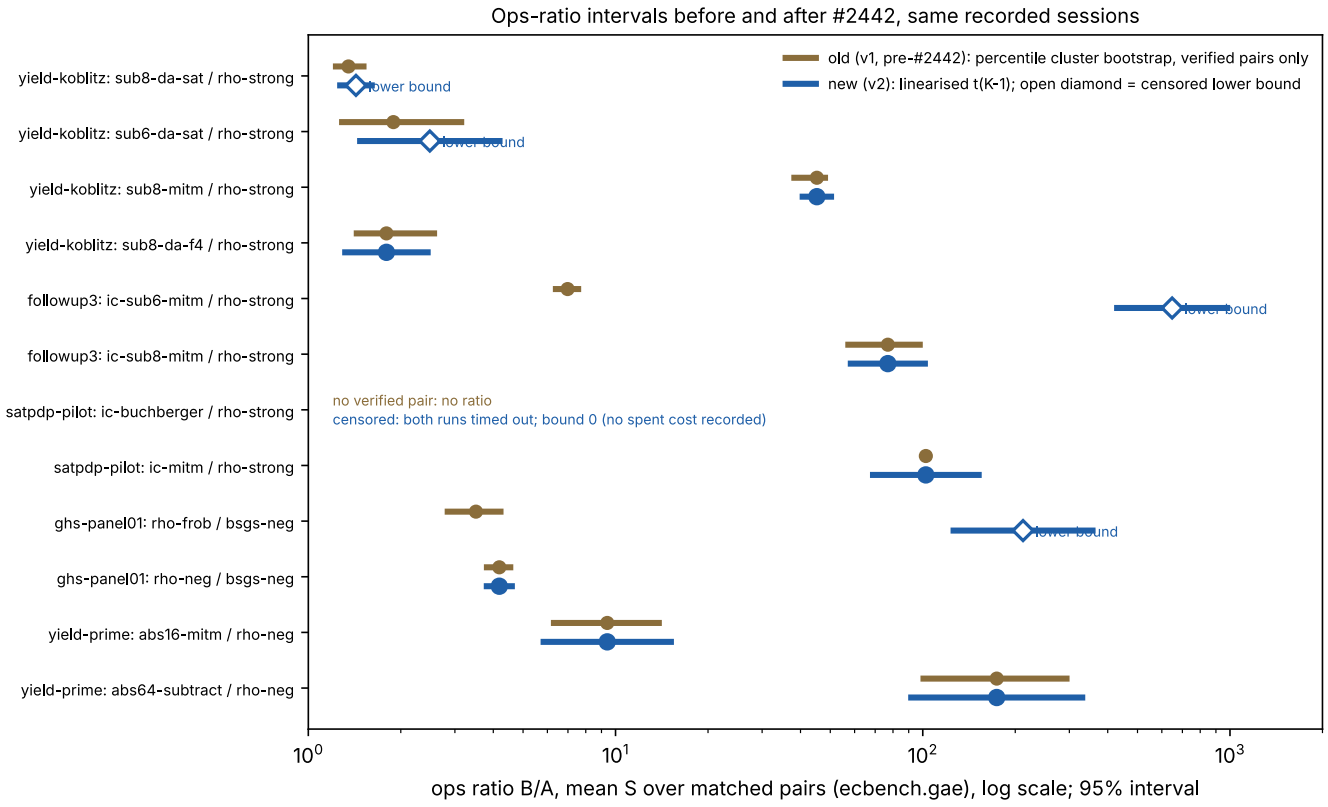
Heavy-tail flag (largest run > 25% of an arm's total S). Counting `ecbench table` rows of the five sessions (`data/tables/new.*.md`): with 4 verified runs per row 71 of 71 rows are flagged, with 2 runs 6 of 6, with 8 runs 1 of 12, with 24 runs 1 of 12. With n runs the largest share is at least 1/n, so at n <= 4 the flag fires on every row and carries no information. In `compare`, the arm-level flag fires only in the K = 2 pilot (top-run share 0.517 with 2 runs); none of the 16-40-workload arms is flagged (top shares 0.03 to 0.22). Recommendation, not implemented here: make the flag's threshold depend on n (or require n >= 8). `survivors only` flags: 3 rows in ghs panel-01 (those with unverified runs); the old table printed their means with no mark. `table` per-row intervals are now t(n-1) and marked `indicative` below five runs (for example rho-frob f2m15: old [2.605, 7.645], new [1.758, 10.635]).

Limits and gates

- Sessions are historical; wall comparisons here are as isolated as the sessions were (levels L0-L2 in the tables). No new measurement, no speedup, no matched-rho ranking is claimed. Rows that bound a ratio are bounds, not coverage statements.
- Validation gate (`docs/VALIDATION_GATE.md`): it governs rows in `benchmarks/README.md` and the IC leaderboard. This study is a methodology comparison, produces no leaderboard row and no benchmark speed row, so no `VALIDATION.json` is created and `benchmarks/README.md` gets a documentary note (outside the admitted table), like the other unadmitted entries. `scripts/validation_gate.py check` is run in the commit.
- Not verified: the PR's withdrawn yield_sweep bound; the wall-jitter simulation; the full-session status of every comparison beyond `admissible: true`.
- Conductor was not available in this sandbox (hooks warn and allow edits); the source change is one `#[ignore]` test in `stats.rs`.

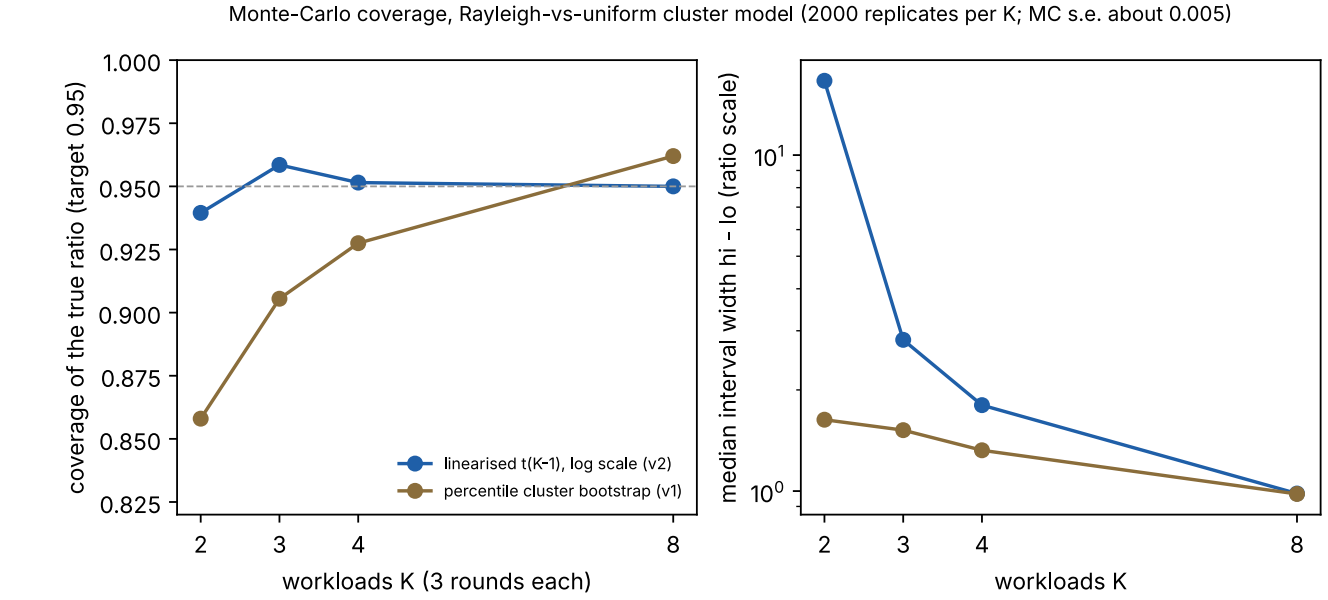
Files: `extract.sh`, `plot.py` (layout only), `data/`, `figures/*.{svg,png}`, `REPORT.typ`, `REPORT.pdf`.

Figures



Source: data/comparisons.tsv (ecbench compare --json at 7edc35a9fb^1 and at the study base). Sessions: yield sweep koblitz = ecbench_yield_sweep_20261004/sessions/koblitz (ECBS1h0d1ce1d04e54); followup3 = ecbench_all_candidates_20261003/sessions/koblitz-followup-3; ghs = ghs_curve_comparison_20261009 panel-01; sat-pilot = ecbench_sat_pdp_20261009 pilot

Figure 1: Fig 1. Ops-ratio intervals, old v1 vs new v2, same sessions. Source data/comparisons.tsv.



Source: data/coverage.tsv from `cargo test --release --lib stats::tests::coverage_table_for_report -- --ignored` (data/stats_tests.log); bootstrap uses 400 resamples.

Figure 2: Fig 2. Coverage and median width, 2000 replicates per K. Source data/coverage.tsv.

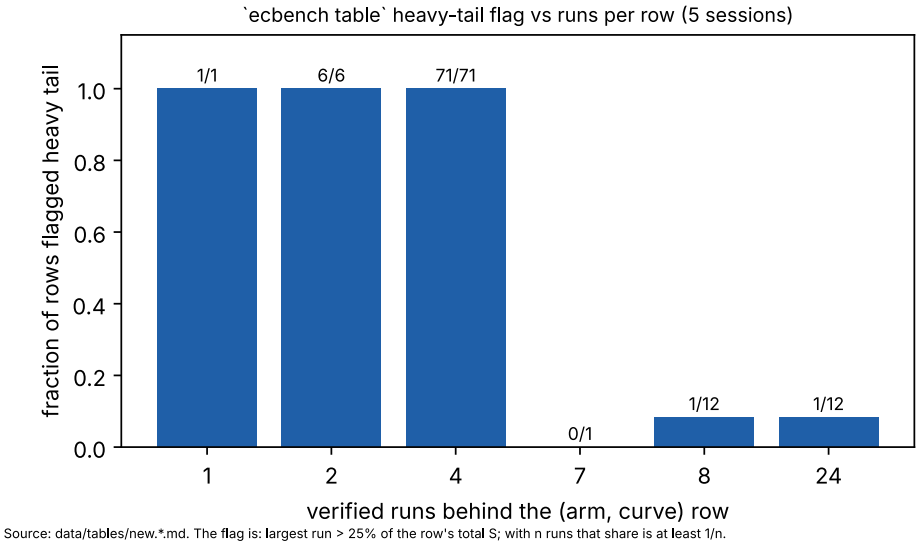


Figure 3: Fig 3. Heavy-tail flag rate against verified runs per table row. Source data/tables.

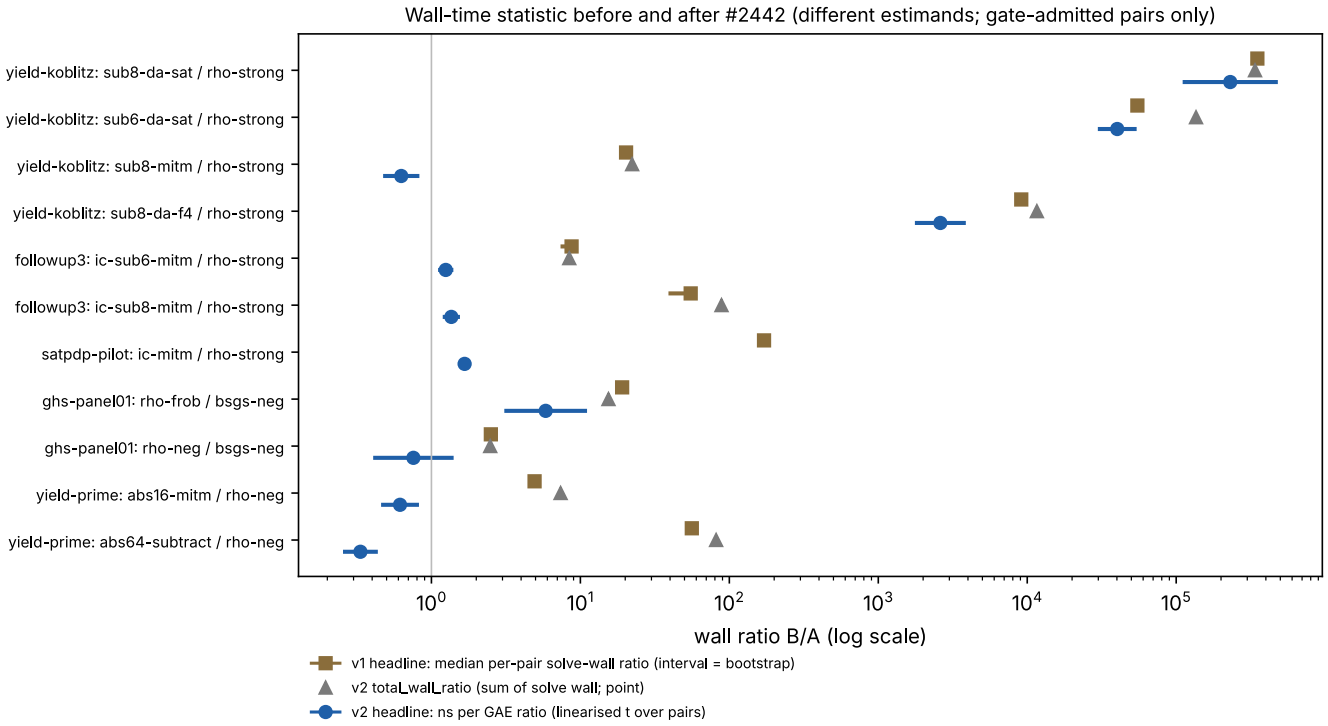


Figure 4: Fig 4. Wall statistic, v1 median per-pair ratio vs v2 ns/GAE ratio. Source data/comparisons.tsv.